

TrustChain-WSN — Students sathi Marathi

Explanation

Event-Driven Lightweight Blockchain-Based Secure Routing for Industrial WSN · Pratyek number project chya code madhun ghetla aahe (`backend/app/core/config.py` , `sim/network.py`)

Ya PDF madhil prashna:

1. Attack chalu astana va nantar Packet Delivery (Graph)

2. 0.40 high value ka ghetli?

3. Throughput chi range

4. Blockchain Ledger Panel

5. Blockchain Configuration Panel

6. Node recover vhayla kiti vel lagto?
7. 100% recovery ka disat nahi?

8. Recovery sathi konta algorithm?

9. Energy vs Control Overhead (Performance Panel)

10. Confidence (Detection Panel) — positive ki negative?

11. Node Inspector — Benign, Battery, MAC/IP

12. Viva sathi ekach olitil uttare

Aadhi mool concept (2 olit): Pratyek node la ek **Trust Score** (0 te 1) asto. Node ne packets drop kele, delay vadhavla kiva khoti identity vaparli ki trust kami hoto. Trust **0.40** chya khali gela ki node **isolate** hoto, routing tyala talun navin path banavte, ani hi ghatna **blockchain block** madhe kayamchi record hote. Simulation dar **2 second la ek round (tick)** chalavte.

1. Attack chalu astana va nantar Packet Delivery — Graph

PDR (Packet Delivery Ratio) mhanje pathavlelya packets paiki kiti packets kharokhar pohochle:

$$\text{PDR} = \text{forwarded} \div (\text{forwarded} + \text{dropped}) \times 100$$

Ha PDR **shevatchya 6 rounds** var mojla jato (rolling window). Purna simulation cha average ghetla asta tar navin attack cha parinam graph var disalach nasta — mhanun fakt latest 6 rounds.

Performance page varil "Before vs After" Bar Graph — 3 bars

Bar	Kay dakhavto	Sadharan value
Before attack	Attack suru honyachya aadhichya round cha PDR (clean network)	Sumare 97–99%
Under attack	Attack chalu astana alela sarvat kami PDR (worst point)	Attackers chya sankhyevar avalambun — jitke jast attackers titka motha dip
After recovery	Sagle nodes parat service madhe alyavarcha PDR	Parat 97–99% chya javal

He tinhi snapshots backend **automatically** ghetu (`live_runner.py` → `_capture_phases()`), konihi button dabaycha garaj nahi.

"PDR Over Time" Line Graph

- Graph madhil **shaded (rangit) bhag** = jya kalat network var attack chalu hota.
- Attack suru → PDR khali jato (malicious node packets fekun deto).
- Trust engine node isolate karto → to routing madhun baher → PDR parat var yeto.

Students la nehmi padnara prashna: "Dip itka chhota ka?" — karan 24 paiki ekach node malicious asel tar baki 22–23 nodes vyavasthit kaam kartat, ani attacker la sahse **1–2 rounds** madhech isolate kela jato. Mhanje fast detection mulech dip chhota rahto — hach project cha fayda aahe. Motha dip dakhvaycha asel tar ekach veli 3–4 nodes var attack inject kara.

2. 0.40 hich value ka ghetli?

`TRUST_THRESHOLD = 0.40` — yapeksha kami trust asnara node isolate hoto. Hi value ashi nivadli aahe ki **changle nodes chukun pakadle jau naye** ani **vait nodes lavkar pakadle jave**.

Threshold	Kay hoil
Khup kami (e.g. 0.10)	Attacker la pakdayla khup rounds lagtil → tovar to packets nasht karat rahil (slow detection)
Khup jast (e.g. 0.70)	Wireless madhe changle nodes suddha kadhi kadhi 0–5% packets gamavtat → ase pramanik nodes chukun isolate hotil (False Positive)
0.40 (aapli value)	Changlya nodes cha trust 0.85–0.99 pasun suru hoto ani javalpas tasach rahto — 0.40 pasun khup lamb. Attackers matra 1–2 rounds madhe 0.40 chya khali jatat. Donhinmadhe clear gap.

Sopya bhashet: 0.40 mhanje "ha node 50% peksha kami vishwasarha aahe" — mhanje to changlya peksha jast vait vagtoy. Trust-based WSN research madhe threshold sadharan 0.3–0.5 madhech ghetla jato; 0.40 ha tya range cha madhya aahe.

Dashboard varche trust groups suddha yachyashi jodlele aahet: **Trusted ≥ 0.70 , Suspect 0.40–0.70, Below threshold < 0.40 .**

Lakshat theva — code madhe ajun ek "0.4" aahe: `network.py` madhe `rng.random() < 0.4` pan aahe. Yacha artha: jo malicious node aadhich threshold chya khali aahe pan isolate zala nahi (he fakt "Existing System" mode madhe hoto), tyacha detection dar round la nahi tar **40% chance ne** log hoto — jenekarun detection table ekach node ne bharun jau naye. Ha 0.4 trust threshold peksha vegla aahe.

3. Throughput chi range

Throughput (kbps) = $180 + (\text{Active nodes} \times 5) + \text{random}(0 \text{ te } 40)$

Paristhiti	Active nodes	Throughput range
24 nodes, sagle active	24	300 – 340 kbps
24 nodes, 2 isolated	22	290 – 330 kbps
Settings madhil minimum (8 nodes)	8	220 – 260 kbps
Settings madhil maximum (60 nodes)	60	480 – 520 kbps

- **180** = base capacity (Sink / backbone).
- **Pratyek active node +5 kbps** deto. Isolate zalela node kahich forward karat nahi → throughput 5 ne kami.
- **random 0–40** = wireless madhil naisargik chadh-utar (interference, traffic badalne). Mhanun graph thoda var-khali hot rahto.

Mhanje attack chya veli node isolate zala ki throughput thoda kami hoto, ani node recover zalyavar parat vadhto.

Pramanik mudda: Throughput ha pratyek packet mojun kadhlela nahi — to varchya formula ne **model** kelela (simulated estimate) aahe. PDR matra kharokhar forwarded/dropped packets mojun kadhla jato. Examiner ne vicharla tar asech sanga.

4. Blockchain Ledger Panel

Blockchain Ledger page var pratyek card = ek **block**. Block fakt **attack detect zalyavarach** banto (Event-Driven). Eka round madhe kitihi detections zale tari **ekach block**; kahich detect zala nahi tar **block nahi**.

Panel varil field	Artha
#index	Block cha number. #0 = Genesis (pahila block, network suru zalyavar)
trigger (genesis / attack)	Block kashamule banla
hash	Tya block cha SHA-256 "fingerprint"
prev	Magchya block cha hash — yanech blocks sakhli sarkhe jodle jatat
merkle	Tya block madhil saglya detection events cha ekatrit hash. Ekhada event badalla tari merkle badalto
Nonce	Mining kartana vaparleli sankhya — hash chi suruvat "00" ne hoi paryant ti vadhavli jate (Proof-of-Work)
Diff	Difficulty = 2 → hash chya suruvatila 2 zero. Muddam kami thevli aahe — sensor nodes kade battery kami aste (Lightweight)
Events	Tya block madhe kiti detections record zale
Sealer	Block koni seal kela — N-001 mhanje Sink node
Chain Verified / Chain Broken	Pratyek block cha prev magchya block chya hash shi julto ka, ani pratyek hash parat calculate kelyavar tasach yeto ka — he check kela jata. Koni juna record badalla tar lagech "Chain Broken"

5. Blockchain Configuration Panel (Settings page)

Option	Artha
Trigger Mode: Event-Driven (aapla project)	Attack detect zala tarach block banto → kami computation, kami energy, kami overhead
Periodic	Thravik velenantar (e.g. dar 10 second la) block — kahi ghadla nasla tari. Energy vaya jate
Continuous	Pratyek transaction / packet sathi block — sarvat jast overhead; sensor network sathi ayogya
Ledger Retention (days)	Records kiti divas thevayche (default 90 divas)
Consensus (Lightweight PoW / PoA / PBFT)	Block valid aahe yavar ekmat kase karaycha. Aapan Lightweight PoW (difficulty 2) vaparto

Periodic ani Continuous he options **comparison sathi** dakhavle aahe — "aamcha Event-Driven mode ya donhi peksha halka aahe" ha project cha main novelty point aahe.

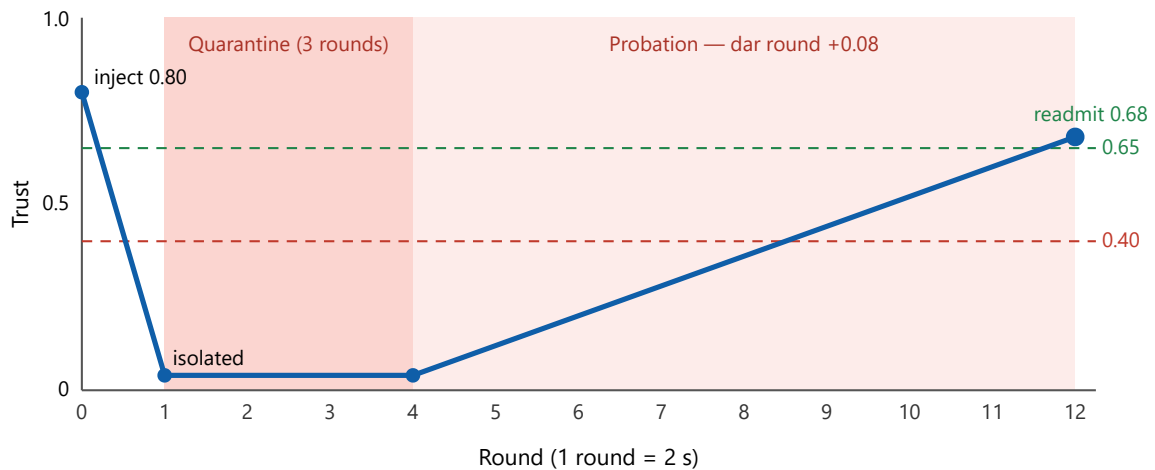
Demo kartana mahatvache: Sadhyachya version madhe Settings page varche he configuration options (Trigger Mode, Retention, Consensus, Trust Threshold slider, Adaptive Threshold) **fakt screen var distat** — "Save" dabalyavar te backend la pathavle jat nahit. Backend nehmi Event-Driven, difficulty 2 ani threshold 0.40 hech vaparto (`config.py`). Examiner ne slider halavun kahi badalat nahi ase vicharla tar: "he real deployment sathiche configuration presets aahet; demo madhe values `config.py` madhe fixed aahet" ase sanga. (Tick speed, Deployed Nodes, Auto-attack, Recover All, Rebuild Network he buttons matra kharokhar kaam kartat.)

6. Node recover vhayla kiti vel lagto?

Config madhil values: `QUARANTINE_TICKS = 3` , `TRUST_REBUILD_RATE = 0.08` , `READMIT_MARGIN = 0.25` , 1 round = 2 second (1x speed).

Node parat routing madhe yenyasathi trust $0.40 + 0.25 = 0.65$ paryant pohochla pahije.

Udaharan — Blackhole attack (trust 0.80 pasun suruvat)



Tappa (stage)	Rounds	Vel (1x speed)
Detect + Isolate (trust 0.80 → 0.04)	1	≈ 2 s
Quarantine — attack behaviour saaf (remediation)	3	≈ 6 s
Probation — trust 0.04 → 0.68	8	≈ 16 s
Injection pasun parat service madhe	12	≈ 24 s

Attack nusar andaje vel

Attack	Drop %	Detect honyas	Ekun recovery (1x)	Ka?
Blackhole	95%	1 round (≈2 s)	≈ 24 s	Lagech pakadla jato, pan trust javaljaval 0 zalyane parat 0.65 gathayla jast vel
Grayhole	50% + delay	1 round	≈ 16–20 s	Trust ~0.2–0.3 paryantach padto → rebuild lavkar
Wormhole	30% + jast delay	1–2 rounds	≈ 18–26 s	Delay penalty mule trust barach khali jau shakto
Sybil	20% + fake identity	2 rounds (≈4 s)	≈ 20 s	Drop kami, pan identity penalty (+0.06) mule pakadla jato

He andaj aahet — random values mule pratyek run madhe thoda farak padto. **Khare mojlele numbers** Performance page varil *Recovery Time by Attack Type* ani *Recovery Scoreboard* (Fastest / Slowest) madhe distat. Speed 4x kela (0.5 s/round) tar hich vel ~4 pat kami hote.

Quarantine / recovery madhe kiti energy (battery) lagte?

Isolated node packets forward karat nahi, pan tyachi battery purna band nasate. Recovery che pratyek stage battery ghete:

Stage	Battery kharch	Ka?
Quarantine (3 rounds)	0.05% / round → 0.15%	Radio fakt listen-only mode madhe
Remediation (ekda)	1.5%	Firmware re-flash + navin security keys
Probation (dar round)	0.15% / round	Watchdog heartbeat probes — node vyavasthit vagtoy ka te tapasne
Blackhole udaharan (3 + 8 rounds)	0.15 + 1.5 + 1.2 = 2.85%	Vel ≈ 22 s (11 rounds × 2 s)

Node isolate hotach system **aadhich sangto** ki recovery la kiti vel ani kiti battery lagel (Node Inspector: *Time to readmission, Battery still needed*). Hi prediction exact aste — test madhe predicted 2.85% ani actual 2.85% hech aale. Recover zalyavar *Battery at isolation → recovery* (e.g. 73.7% → 70.9%) ani *Recovery energy cost* pan disto.

Existing System (baseline mode): ithe recovery hotach nahi — operator ne manually restore karepryant node compromised rahto. Yach comparison madhun aaplya system cha fayda disto.

7. 100% recovery ka disat nahi?

"Recovery" che 4 vegvegle artha aahet — pratyekache karan vegla:

Kay 100% hot nahi	Karan
PDR 100% var parat yet nahi (97–99% var thambto)	Attack nastana suddha pramanik nodes wireless noise mule 0–5% packets gamavtat. Mhanun attack chya <i>aadhi</i> suddha PDR 100% navhta. "After" bar chi tulana "Before" bar shi kara, 100% shi nahi. Shivay PDR 6 rounds cha window aahe — attack che rounds window madhun baher jaiparyant (~12 s) PDR halu halu var yeto.
Trust 1.0 paryant parat jat nahi	Node parat alyavar trust fakt ~0.65–0.72 asto. Ekda attack kelelya node var lagech purna vishwas thevne dhokadayak — trust halu halu kamavava lagto (real life madhil credit score sarkha).
Energy parat yet nahi	Battery recharge hot nahi — kharch zaleli energy kayamchi geli. Ulat recovery swatah suddha sumare 2.7–2.9% battery ghetle (quarantine + scrub + probation).
Kahi incidents "Unresolved"	(a) Node ajun quarantine/probation madhe aahe — thoda vel thamba. (b) Existing System mode chalu aahe — tithe auto-recovery band aste.

0.40 var nahi tar 0.65 var parat ka ghetu? (Hysteresis) Jar node la 0.40 varach parat ghetla, tar ekach wait round ani to parat isolate — node satat aat-baher karat rahil ("flapping"). 0.25 cha gap thevlyamule recovered node stable rahto.

8. Recovery sathi konta Algorithm?

Project madhe 4 algorithms aahet. Tyanchi **standard (literature madhli) naave** — viva madhe high naave sanga:

Kaam	Algorithm che naav	Reference (paper)
Attack detection	Weighted Trust Evaluation (WTE)	Atakli et al., 2008 — "Malicious Node Detection in WSN using Weighted Trust Evaluation"
Secure routing	Trust-Aware AODV (TA-AODV)	AODV — Perkins & Royer, 1999
Node recovery	Second-Chance Redemption (hysteresis sobat)	CONFIDANT protocol — Buchegger & Le Boudec, 2002
Blockchain ledger	Event-Triggered Proof-of-Work (SHA-256 + Merkle tree)	Nakamoto, 2008

Recovery algorithm = Second-Chance Redemption: isolated node la "dusri sandhi" milte — to quarantine madhe saaf kela jato, nantar probation var trust parat kamavto, ani threshold peksha jast (0.65) trust zala tarach parat ghetla jato. Code: `network.py` → `_probation_tick()`. Dashboard varchya Automatic Recovery banner madhe hi sagli naave disat.

1. **Detection:** dar round la trust penalty:

```
penalty = (dropped/traffic) × 0.08 × 10 + max(0, delay - 1) × 0.05 + 0.06 (fakt Sybil)
```

Trust 0.40 chya khali → node **isolate**.

- 2. **Rerouting (Trust-Aware AODV):** pudhcha hop asa nivadla jato ki $\text{distance-to-sink} \div \text{trust}$ sarvat kami asel. Isolated / kami trust vale nodes candidate list madhunach kadhle jatat. Tyamule data lagech surakshit margane jato.
- 3. **Quarantine + Remediation:** Node 3 rounds vegla thevla jato, nantar tyacha attack behaviour automatically saaf (malicious flag, attack, wormhole tunnel kadhle).
- 4. **Probation:** dar round trust +0.08. Trust ≥ 0.65 zalyavarach node parat routing pool madhe. Fakt vel gelyane parat yet nahi — trust prove karava lagto.
- 5. **Blockchain log:** Detection chi nond block madhe. Pratyek stage chi vel (detect / isolate / remediate / recover) mojli jate.

Manual option: Node Inspector madhil *Restore Now* button pipeline skip karun node lagech parat aanto (trust 0.85). Scoreboard madhe he "Manual" mhanun mojla jata.

9. Energy vs Control Overhead (Performance Panel)

Ekach graph madhe don lines:

Line	Artha	Kasa badalto
Avg energy % (green area)	Active nodes chi average battery	Pratyek node chi suruvat 40–99% battery ne; packets forward kartana dar round 0.1–0.6% kami. Tyamule halu halu khali janari line.
Overhead % (orange line)	Control overhead = data packets shivay network la pathvave lagnare extra control messages (routing updates, blockchain blocks)	$2.0 + (\text{ekun detections} \times 0.05)$ — fakt detection zalyavarach thoda vadhto, baki vel flat.

Ha graph kay prove karto? Overhead line **flat** rahte karan blockchain timer var nahi tar fakt attack var block banavto (Event-Driven). Periodic / Continuous blockchain madhe hi line satat var jat rahili asti ani energy jast vegane sampli asti.

"Control" cha artha: ithe Control mhanje "control traffic / control messages" — Settings madhil "control" buttons shi yacha sambandh nahi.

Energy line kadhi achanak udi ka marte? Average fakt *active* nodes chi ghetli jate. Ekhada node isolate zala ki to average madhun baher jato (ani parat alyavar parat yeto) — tyamule average thoda var kiva khali udi maru shakto. Isolated node forward karat nahi, mhanun tyachi battery khup halu (0.05–0.15% / round) kharch hote — normal forwarding peksha kami.

Throughput pramanech Overhead suddha model kelela (formula) number aahe, pratyaksha messages mojun kadhlela nahi.

10. Confidence (Detection Panel) — positive ki negative?

$\text{Confidence} = 0.70 + (\text{drop ratio} \times 0.30)$ (maximum 0.99)

Confidence mhanje "ha node kharokhar attacker aahe yachi system la kiti khatri aahe". Jitke jast packets drop, titki khatri jast.

Attack	Drop ratio	Confidence
Blackhole	~95%	~98–99%
Grayhole	~50%	~85%
Wormhole	~30%	~79%
Sybil	~20%	~76%
Chukun khali gelela changla node	0–5%	~70%

Positive ki Negative?

- **Confidence kadhihi negative nasto** — to nehmi 70% te 99% madhe asto. 70% = "trust threshold olandla mhanun flag kela, pan drops kami aahet"; 99% = "javaljaval nakki attacker".
- **Trust cha badal negative asto:** row expand kelyavar "Trust before → after" disto, e.g. 0.800 → 0.040 . Ha badal nehmi khalchya dishene (negative) — mhanje penalty lagli.
- **True Positive / False Positive:** khara attacker pakadla = True Positive (confidence jast, 76–99%). Changla node chukun pakadla = False Positive (confidence ~70%, sarvat kami). Mhanun **kami confidence = jast kaljipurvak check kara**. 0.40 threshold mule False Positive khup kvachit hotat.

Detection table madhil itar columns: **Severity** (Blackhole/Wormhole = Critical, Sybil = High, Grayhole = Medium), **Status** (Isolated = quarantine kela; Detected = fakt penalty, nigrani khali), **Block #** = hi nond konatya blockchain block madhe aahe.

11. Network Map — Node Inspector Panel

Map varchya konatyahi node var click kelyavar ujkade he panel ughadte:

Field	Artha
MAC address	Pratyek node cha unique hardware ID, e.g. <code>02:57:53:4E:00:05</code> . "02" = locally administered, "57:53:4E" = ASCII madhe "WSN", shevatche 2 bytes = node number.
Static IP	Deployment veli dilela fixed IP: <code>10.10.0.<node number></code> . Sink = 10.10.0.1, N-005 = 10.10.0.5. Map var node var hover kela tar pan disto.
Phase chip	Active → Compromised → Detected → Isolated → Remediating → Recovered — node sadhya konatya stage madhe aahe
Area	Node konatya Zone (A/B/C) madhe — javalchya Cluster Head nusar
Trust score	0–1; green = vishwasarha, red = dhokadayak
Battery (lifeline)	Tya node chi urleli battery % — bar sobat (green > 50%, orange 20–50%, red < 20%). Khali " Lifetime ≈ X left " = sadhya chya kharcha rate ne battery kiti vel tikel (battery ÷ % per round × round time). Suruvat 40–99%, forward kartana dar round 0.1–0.6% kami. Recharge hot nahi. Map var pratyek node khali pan chhota battery bar disto.
Packets forwarded / dropped	Suruvatipasun ekun pudhe pathavlele / takun dilele packets. Dropped 50 peksha jast asel tar red disto.
Ground truth: Benign	Benign = nirupadravi / changla node — to kharokhar attacker nahi. Malicious asel tar ithe attack che naav (Blackhole, Sybil...) red rangat disto.
Wormhole partner	Wormhole don nodes milun kartat (tunnel) — dusra sathidar node
Incident timing	Detected after / Isolated after / Back in service after / End to end — tya node chya attack chi mojleli vel
Recovering automatically (progress bar)	Attack saaf zala, aata probation chalu — bar 100% (trust 0.65) zalyavar node parat. Sobat: <i>Time to readmission, Battery spent so far, Battery still needed, Total recovery cost.</i> Quarantine stage madhe he box orange rangat "In quarantine — being scrubbed" dakhavte.

"Ground truth" ka dakhavto? Simulation la mahit asta ki konta node kharokhar malicious aahe. Real network madhe he mahit nasta — system la fakt trust score disto. Ground truth ani trust engine cha nirnay yanchi tulana karun aapan check karu shakto ki detection **barobar** hota ka. E.g. Ground truth "Benign" pan trust red → ha False Positive.

Mahatvache: Attack remediate zalyanantar (Isolated pan "Recovering automatically" distana) Ground truth "Benign" dakhavto — karan attack behaviour aadhich saaf zala aahe; node fakt trust parat kamavtoy.

12. Viva sathi ekach olitil uttare

Prashna	Uttar
0.40 ka?	Changle nodes (0.85–0.99) ani attackers (1–2 rounds madhe khali) yanchyat clear gap; false positive kami, detection fast.
Readmit 0.65 ka?	Hysteresis — node aat-baher flap hou naye mhanun $0.40 + 0.25$ margin.
Recovery vel?	1x speed la sadharan 16–26 s (Blackhole ≈ 24 s). Existing system madhe recovery nahich.
Algorithms che naav?	Detection: Weighted Trust Evaluation (WTE) · Routing: Trust-Aware AODV (TA-AODV) · Recovery: Second-Chance Redemption (CONFIDANT) · Ledger: Event-Triggered PoW.
Recovery kasa hoto?	Quarantine (3 rounds) → Remediation → Probation (+0.08/round) → Readmit ≥ 0.65 ; TA-AODV ne reroute.
Recovery la battery kiti?	Sumare 2.7–2.9% (0.05%/round quarantine + 1.5% scrub + 0.15%/round probation).
MAC / IP?	MAC 02:57:53:4E:00:<n>, static IP 10.10.0.<n> — Node Inspector madhe.
PDR 100% ka nahi?	Pramanik nodes suddha 0–5% packets wireless noise mule gamavtat; tulana "Before" shi kara.
Throughput range?	$180 + 5 \times \text{active nodes} + 0-40 \rightarrow 24$ nodes la 300–340 kbps.
Overhead flat ka?	Event-Driven blockchain — block fakt attack var, timer var nahi.
Confidence?	$0.70 + \text{drop ratio} \times 0.30$; nehmi positive (70–99%). Trust cha badal negative.
Benign?	Kharokhar changla (attacker nasnara) node — simulation cha ground truth.
Block kadhi banto?	Fakt detection zalelya round madhe, ek round = jastit jast ek block; SHA-256 PoW, difficulty 2.

TrustChain-WSN (bc-routing) · Sagli values: TRUST_THRESHOLD 0.40, READMIT_MARGIN 0.25, QUARANTINE_TICKS 3, TRUST_REBUILD_RATE 0.08, DROP_PENALTY 0.08, DELAY_PENALTY 0.05, BLOCK_DIFFICULTY 2, QUARANTINE_IDLE_DRAIN 0.05, REMEDIATION_ENERGY 1.5, PROBATION_DRAIN 0.15, tick = 2 s (1x). Recovery vela andaje aahet; achuk mojleli vel dashboard var live diste.